Task 1: Data Acquisition

```
# Load the libraries

library(tidyverse)
```

```
── Attaching core tidyverse packages ────────────────────────
tidyverse 2.0.0 ──
✔ dplyr     1.2.0     ✔ readr     2.2.0
✔ forcats   1.0.1     ✔ stringr   1.6.0
✔ ggplot2   4.0.2     ✔ tibble    3.3.1
✔ lubridate 1.9.4     ✔ tidyr     1.3.2
✔ purrr     1.2.1
── Conflicts ────────────────────────────────────────
tidyverse_conflicts() ──
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()    masks stats::lag()
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to
force all conflicts to become errors
```

```
library(glue)
library(readxl)
library(httr2)
library(gt)
library(ggplot2)
library(scales)
```

```
Attaching package: 'scales'

The following object is masked from 'package:purrr':
```

```
    discard

The following object is masked from 'package:readr':

    col_factor
```

```
# Create load_atus_data function and modify it for tasks 3, 4, 5, a

load_atus_data <- function(file_base = c("resp", "rost", "act")){
    if(!dir.exists(file.path("data", "mp02"))){
        dir.create(file.path("data", "mp02"), showWarnings=FALSE, r
    }

    file_base <- match.arg(file_base)

    file_name_out <- file.path("data",
                               "mp02",
                               glue("atus{file_base}_0324.dat"))

    if(!file.exists(file_name_out)){

        url_end <- glue("atus{file_base}-0324.zip")

        temp_zip <- tempfile(fileext=".zip")
        temp_dir <- tools::file_path_sans_ext(temp_zip)

        request("https://www.bls.gov") |>
            req_url_path("tus", "datafiles", url_end) |>
            req_headers(`User-Agent` = "Mozilla/5.0 (Macintosh; Int
            req_perform(temp_zip)

        unzip(temp_zip, exdir = temp_dir)

        temp_file <- file.path(temp_dir, basename(file_name_out))

        file.copy(temp_file, file_name_out)
```

```
    }

    file <- read_csv(file_name_out,
                     show_col_types = FALSE)

    switch(file_base,
           resp = file |>
                  rename(survey_weight = TUFNWGTP,
                         time_alone = TRTALONE,
                         survey_year = TUYEAR) |>
                  mutate(
                    # Task 3: Number of own household children unde
                    n_children_under18 = TRYHHCHILD,

                    # Retirement status from TERET1: 2 = retired, 1
                    retired = if_else(TERET1 == 2, TRUE, FALSE),

                    # Task 3: Married status based on spouse presen
                    # TRSPPRES: 1 = spouse present, 2 = unmarried p
                    married = case_match(TRSPPRES,
                      1 ~ "Yes",
                      2 ~ "No",
                      3 ~ "No"
                    ),

                    # Task 3 and task 4: Employment status
                    employment_status = case_match(TELFS,
                      1 ~ "Employed - At Work",
                      2 ~ "Employed - Absent",
                      3 ~ "Unemployed - On Layoff",
                      4 ~ "Unemployed - Looking",
                      5 ~ "Not in Labor Force"
                    ),

                    # Task 3: College enrollment
                    # TESCHLVL: -1 = not enrolled, 1 = high school,
```

```
            # We need enrolled AND at college level (not HS
            college_enrolled = if_else(TESCHLVL == 2, "Yes",

            # Task 5:
            line_no = TULINENO,
            hourly_earnings = TRERNHLY,

            # Task 6:

            # Survey month (1-12)
            survey_month = TUMONTH,

            # Industry code (for identifying education work
            industry_code = TEIO1ICD,
        ),
# rost = file |>
        # mutate(sex = case_match(TESEX, 1 ~ "M", 2 ~ "F"

rost = file |>
        mutate(sex = case_match(TESEX, 1 ~ "M", 2 ~ "F"),
               age = TEAGE,
               line_no = TULINENO),
act = file |>
     mutate(activity_n_of_day = TUACTIVITY_N,
            time_spent_min = TUACTDUR24,
            start_time = TUSTARTTIM,
            stop_time = TUSTOPTIME,
            level1_code = paste0(TRTIER1P, "0000"),
            level2_code = paste0(TRTIER2P, "00"),
            level3_code = TRCODEP,

            # Task 3: Location of activity
            activity_location = case_match(TEWHERE,
              1  ~ "Respondent's Home",
              2  ~ "Respondent's Workplace",
              3  ~ "Someone Else's Home",
```

```
                        4  ~ "Restaurant or Bar",
                        5  ~ "Place of Worship",
                        6  ~ "Grocery Store",
                        7  ~ "Other Store/Mall",
                        8  ~ "School",
                        9  ~ "Outdoors Away from Home",
                        10 ~ "Library",
                        11 ~ "Other Place",
                        12 ~ "Car/Truck/Motorcycle (Driver)",
                        13 ~ "Car/Truck/Motorcycle (Passenger)",
                        14 ~ "Walking",
                        15 ~ "Bus",
                        16 ~ "Subway/Train",
                        17 ~ "Bicycle",
                        18 ~ "Boat/Ferry",
                        19 ~ "Taxi/Rideshare",
                        20 ~ "Airplane",
                        21 ~ "Other Transportation",
                        30 ~ "Bank",
                        31 ~ "Gym/Health Club",
                        32 ~ "Post Office",
                        99 ~ "Unspecified",
                        -1 ~ NA_character_
                      ))
            ) |>
                rename(participant_id = TUCASEID) |>
                select(matches("[:lower:]",
                               ignore.case=FALSE))
}
```

```
# Create load_atus_activities function and modify it for task 3

load_atus_activities <- function(){
    if(!dir.exists(file.path("data", "mp02"))){
        dir.create(file.path("data", "mp02"), showWarnings=FALSE, r
```

```
    }

    dest_file <- file.path("data",
                           "mp02",
                           "atus_activity_codes.csv")

    if(!file.exists(dest_file)){
        download.file("https://michael-weylandt.com/STA9750/mini/at
                      quiet=TRUE,
                      destfile=dest_file,
                      mode="wb")
    }

    read_csv(dest_file, show_col_types = FALSE)
}
```

```
# Acquire the four datasets

resp_df <- load_atus_data("resp")
```

```
Warning: There was 1 warning in `mutate()`.
ℹ In argument: `married = case_match(TRSPPRES, 1 ~ "Yes", 2 ~ "No",
3 ~ "No")`.
Caused by warning:
! `case_match()` was deprecated in dplyr 1.2.0.
ℹ Please use `recode_values()` instead.
```

```
rost_df <- load_atus_data("rost")
act_df <- load_atus_data("act")
activities_df <- load_atus_activities()
```

Look at the basic structure of the acquired datasets

```
glimpse(resp_df)
```

```
Rows: 252,808
Columns: 13
$ participant_id     <dbl> 2.00301e+13, 2.00301e+13, 2.00301e+13,
2.00301e+13,…
$ time_alone         <dbl> 525, 0, 475, 350, 140, 170, 132, 105, 0,
626, 30, 4…
$ survey_year        <dbl> 2003, 2003, 2003, 2003, 2003, 2003,
2003, 2003, 200…
$ survey_weight      <dbl> 8155462.7, 1735322.5, 3830527.5,
6622023.0, 3068387…
$ n_children_under18 <dbl> -1, 0, -1, 9, 14, 2, 9, 14, 3, 4, 4, 7,
14, 17, 2, …
$ retired            <lgl> FALSE, FALSE, FALSE, FALSE, FALSE,
FALSE, FALSE, FA…
$ married            <chr> "Yes", "Yes", "Yes", "Yes", "Yes", "No",
"Yes", "No…
$ employment_status  <chr> "Employed - Absent", "Employed - At
Work", "Employe…
$ college_enrolled   <chr> "No", "No", "No", "No", "No", "Yes",
"No", "No", "N…
$ line_no            <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
1, 1, 1, 1, …
$ hourly_earnings    <dbl> 2200, -1, 1250, -1, -1, -1, -1, 950,
1400, 1200, -1…
$ survey_month       <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
1, 1, 1, 1, …
$ industry_code      <dbl> 7860, 1690, 8470, -1, 7970, 7860, 8470,
8190, 7070,…
```

```
head(resp_df)
```

```
# A tibble: 6 × 13
  participant_id time_alone survey_year survey_weight
n_children_under18 retired
           <dbl>      <dbl>       <dbl>         <dbl>
<dbl> <lgl>
1        2.00e13        525        2003      8155463.
-1 FALSE
2        2.00e13          0        2003      1735323.
0 FALSE
3        2.00e13        475        2003      3830527.
-1 FALSE
4        2.00e13        350        2003      6622023.
9 FALSE
5        2.00e13        140        2003      3068387.
14 FALSE
6        2.00e13        170        2003      3455425.
2 FALSE
# ℹ 7 more variables: married <chr>, employment_status <chr>,
#   college_enrolled <chr>, line_no <dbl>, hourly_earnings <dbl>,
#   survey_month <dbl>, industry_code <dbl>
```

```
glimpse(rost_df)
```

```
Rows: 687,357
Columns: 4
$ participant_id <dbl> 2.00301e+13, 2.00301e+13, 2.00301e+13,
2.00301e+13, 2.0…
$ sex            <chr> "M", "F", "F", "F", "M", "M", "F", "F", "M",
"F", "M", …
$ age            <dbl> 60, 72, 37, 41, 42, 3, 0, 26, 35, 36, 39,
11, 9, 51, 50…
$ line_no        <dbl> 1, 2, 3, 1, 2, 3, 4, 1, 2, 1, 2, 3, 4, 1, 2,
3, 4, 1, 2…
```

```r
head(rost_df)
```

```
# A tibble: 6 × 4
  participant_id sex     age line_no
           <dbl> <chr> <dbl>   <dbl>
1        2.00e13 M        60       1
2        2.00e13 F        72       2
3        2.00e13 F        37       3
4        2.00e13 F        41       1
5        2.00e13 M        42       2
6        2.00e13 M         3       3
```

```r
glimpse(act_df)
```

```
Rows: 4,880,021
Columns: 9
$ participant_id    <dbl> 2.00301e+13, 2.00301e+13, 2.00301e+13,
2.00301e+13, …
$ activity_n_of_day <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 1, 2, 3, 4, 5,
6, 7, 8, 9…
$ time_spent_min    <dbl> 60, 30, 600, 150, 5, 175, 270, 10, 140,
180, 60, 60,…
$ start_time        <time> 04:00:00, 05:00:00, 05:30:00, 15:30:00,
18:00:00, 1…
$ stop_time         <time> 05:00:00, 05:30:00, 15:30:00, 18:00:00,
18:05:00, 2…
$ level1_code       <chr> "130000", "010000", "010000", "120000",
"110000", "1…
$ level2_code       <chr> "130100", "010200", "010100", "120300",
"110100", "1…
$ level3_code       <chr> "130124", "010201", "010101", "120303",
"110101", "1…
$ activity_location <chr> "Outdoors Away from Home", NA, NA,
"Respondent's Hom…
```

```r
head(act_df)
```

```
# A tibble: 6 × 9
  participant_id activity_n_of_day time_spent_min start_time
stop_time
           <dbl>             <dbl>          <dbl> <time>     
<time>
1        2.00e13                 1             60 04:00      
05:00
2        2.00e13                 2             30 05:00      
05:30
3        2.00e13                 3            600 05:30      
15:30
4        2.00e13                 4            150 15:30      
18:00
5        2.00e13                 5              5 18:00      
18:05
6        2.00e13                 6            175 18:05      
21:00
# ℹ 4 more variables: level1_code <chr>, level2_code <chr>,
level3_code <chr>,
#   activity_location <chr>
```

```r
glimpse(activities_df)
```

```
Rows: 477
Columns: 6
$ code_level1 <chr> "010000", "010000", "010000", "010000",
"010000", "010000"…
$ code_level2 <chr> "010100", "010100", "010100", "010200",
"010200", "010300"…
$ code_level3 <chr> "010101", "010102", "010199", "010201",
"010299", "010301"…
$ task_level1 <chr> "Personal Care", "Personal Care", "Personal
Care", "Person…
$ task_level2 <chr> "Sleeping", "Sleeping", "Sleeping", "Grooming",
"Grooming"…
$ task_level3 <chr> "Sleeping", "Sleeplessness", "Other Sleeping",
"Washing, d…
```

```r
head(activities_df)
```

```
# A tibble: 6 × 6
  code_level1 code_level2 code_level3 task_level1   task_level2
task_level3
  <chr>       <chr>       <chr>       <chr>         <chr>
<chr>
1 010000      010100      010101      Personal Care Sleeping
Sleeping
2 010000      010100      010102      Personal Care Sleeping
Sleeplessn…
3 010000      010100      010199      Personal Care Sleeping
Other Slee…
4 010000      010200      010201      Personal Care Grooming
Washing, d…
5 010000      010200      010299      Personal Care Grooming
Other Groo…
6 010000      010300      010301      Personal Care Health-Related
… Health-rel…
```

Task 3: Adding Additional Demographic Variables

```
# Check the new Task 3 demographic variables that were added in the

resp_df |> count(married)
```

```
# A tibble: 2 × 2
  married      n
  <chr>    <int>
1 No      126381
2 Yes     126427
```

```
resp_df |> count(employment_status)
```

```
# A tibble: 5 × 2
  employment_status           n
  <chr>                   <int>
1 Employed - Absent        6824
2 Employed - At Work     148458
3 Not in Labor Force      86841
4 Unemployed - Looking     9501
5 Unemployed - On Layoff   1184
```

```
resp_df |> count(college_enrolled)
```

```
# A tibble: 2 × 2
  college_enrolled      n
  <chr>             <int>
1 No               239914
2 Yes               12894
```

```
resp_df |> summarise(min = min(n_children_under18, na.rm = TRUE),
                     max = max(n_children_under18, na.rm = TRUE),
                     mean = mean(n_children_under18, na.rm = TRUE))
```

```
# A tibble: 1 × 3
    min   max  mean
  <dbl> <dbl> <dbl>
1    -1    17  2.63
```

```
act_df |> count(activity_location, sort = TRUE)
```

```
# A tibble: 26 × 2
   activity_location                      n
   <chr>                            <int>
 1 Respondent's Home               2095267
 2 <NA>                             921554
 3 Car/Truck/Motorcycle (Driver)    672364
```

```
 4 Respondent's Workplace            231498
 5 Someone Else's Home               161397
 6 Car/Truck/Motorcycle (Passenger)  153114
 7 Other Place                       146333
 8 Other Store/Mall                   97664
 9 Restaurant or Bar                  92900
10 Walking                            65016
# ℹ 16 more rows
```

Task 4: Exploratory Questions - Inline Values

```
# Question 1: How many unique respondents since 2003?

n_respondents <- n_distinct(resp_df$participant_id)
```

```
# Question 2: Americans enjoy watching a variety of sports. How many

# Attending Sports or Recreational Events is the Level 2 category
# Each Level 3 entry under it is a specific sport
n_sports_watching <- activities_df |>
  filter(task_level2 == "Attending Sports or Recreational Events")
  filter(!is.na(task_level3)) |>
  nrow()
```

```
# Question 3: Approximately what percent of Americans are retired?

# TERET1: 2 = retired, 1 = not retired, negative = not applicable
# People not asked are treated as not retired
# Weighted mean of boolean gives the population-weighted percentage
pct_retired <- resp_df |>
  summarise(pct = weighted.mean(retired, survey_weight, na.rm = TRU
```

```
  pull(pct) |>
  round(1)
```

```
# Question 4: On average, how many hours do Americans sleep per nigl

# Sleep = level3_code "010101"; codes are character strings
avg_sleep_hours <- act_df |>
  filter(level3_code == "010101") |>
  group_by(participant_id) |>
  summarise(sleep_min = sum(time_spent_min)) |>
  left_join(resp_df |> select(participant_id, survey_weight),
            by = "participant_id") |>
  summarise(avg = weighted.mean(sleep_min, survey_weight, na.rm = Tl
  pull(avg) |>
  round(2)
```

```
# Question 5: On average, how many hours do Americans with one or m

# 030100 = Caring for & helping HH children
# 030200 = Activities related to HH children's education
# 030300 = Activities related to HH children's health
# right_join ensures parents with 0 childcare minutes are included

avg_childcare_hours <- act_df |>
  filter(level2_code %in% c("030100", "030200", "030300")) |>
  group_by(participant_id) |>
  summarise(care_min = sum(time_spent_min)) |>
  right_join(
    resp_df |>
      filter(n_children_under18 >= 1) |>
      select(participant_id, survey_weight),
    by = "participant_id"
  ) |>
```

```
  mutate(care_min = replace_na(care_min, 0)) |>
  summarise(avg = weighted.mean(care_min, survey_weight, na.rm = TRU
  pull(avg) |>
  round(2)
```

```
# Just looking at each variable before I write my inline code

n_respondents
```

```
[1] 252808
```

```
n_sports_watching
```

```
[1] 33
```

```
pct_retired
```

```
[1] 14.9
```

```
avg_sleep_hours
```

```
[1] 8.71
```

```
avg_childcare_hours
```

```
[1] 0.89
```

Answers

1. Since 2003, 252808 different respondents have answered an ATUS survey.

2. ATUS asks about watching 33 different sports as a Level 3 task.

3. About 14.9% of Americans are retired.

4. On average, Americans sleep 8.71 hours per night.

5. On average, Americans with one or more children in the home spend 0.89 hours per day taking care of their children.

Task 5: Exploratory Questions - Table Answers

```
# Create resp_full from resp_df dataset

resp_full <- resp_df |>
  left_join(
    rost_df |> select(participant_id, line_no, age, sex),
    by = c("participant_id", "line_no")
  )
```

```
# Question 1: Do retired persons spend more time on lawn-care than

# Identify lawn/garden activity codes from the codebook
lawn_codes <- activities_df |>
  filter(str_detect(tolower(task_level2), "lawn|garden")) |>
  pull(code_level3)

# Sum lawn care minutes per respondent, keeping people with 0 minut
q1_table <- act_df |>
  filter(level3_code %in% lawn_codes) |>
  group_by(participant_id) |>
  summarise(lawn_min = sum(time_spent_min)) |>
  right_join(resp_full |> select(participant_id, survey_weight, ret
             by = "participant_id") |>
```

```
  mutate(lawn_min = replace_na(lawn_min, 0)) |>
  # Keep only retired or employed full-time (at work)
  filter(retired == TRUE | employment_status == "Employed - At Work
  mutate(group = if_else(retired, "Retired", "Employed Full-Time"))
  group_by(group) |>
  summarise(
    avg_minutes = round(weighted.mean(lawn_min, survey_weight, na.r
    .groups = "drop"
  )

q1_table |>
  gt() |>
  cols_label(group = "Group", avg_minutes = "Avg. Daily Minutes on 
  tab_header(title = "Time Spent on Lawn Care: Retired vs. Employed
```

Time Spent on Lawn Care: Retired vs. Employed	
Group	Avg. Daily Minutes on Lawn Care
Employed Full-Time	9.1
Retired	23.1

```
# Question 2: How does the amount of time spent on social and recre

# Identify social/leisure and sports/recreation level 3 codes
social_rec_codes <- activities_df |>
  filter(str_detect(tolower(task_level1), "social|leisure|sport|rec
  pull(code_level3)

q2_table <- act_df |>
  filter(level3_code %in% social_rec_codes) |>
```

```
  group_by(participant_id) |>
  summarise(social_min = sum(time_spent_min)) |>
  right_join(resp_full |> select(participant_id, survey_weight, mar
             by = "participant_id") |>
  mutate(social_min = replace_na(social_min, 0)) |>
  group_by(married) |>
  summarise(
    avg_minutes = round(weighted.mean(social_min, survey_weight, na
    .groups = "drop"
  )

q2_table |>
  gt() |>
  cols_label(married = "Married?", avg_minutes = "Avg. Daily Minute
  tab_header(title = "Time Spent on Social & Recreational Activitie
```

Time Spent on Social & Recreational Activities by Marital Status

Married?	Avg. Daily Minutes
No	321.1

```
# Question 3: Where do high-earning Americans spend the majority of

# Define high-earning as top 25% of positive earners
earnings_cutoff <- resp_full |>
  filter(hourly_earnings > 0) |>
  pull(hourly_earnings) |>
  quantile(0.75)
```

```
high_earner_ids <- resp_full |>
  filter(hourly_earnings >= earnings_cutoff) |>
  select(participant_id, survey_weight)

q3_table <- act_df |>
  inner_join(high_earner_ids, by = "participant_id") |>
  filter(!is.na(activity_location)) |>
  group_by(activity_location) |>
  summarise(
    total_weighted_min = sum(time_spent_min * survey_weight),
    .groups = "drop"
  ) |>
  mutate(pct_time = round(total_weighted_min / sum(total_weighted_m
  arrange(desc(pct_time)) |>
  head(10)

q3_table |>
  select(activity_location, pct_time) |>
  gt() |>
  cols_label(activity_location = "Location", pct_time = "% of Time"
  tab_header(title = "Where High-Earning Americans Spend Their Time
             subtitle = paste0("Top 25% of earners (>= $",
                               round(earnings_cutoff / 100, 2), "/h
```

Where High-Earning Americans Spend Their Time

Top 25% of earners (>= $20.92/hour)

Location	% of Time
Respondent's Home	42.6
Respondent's Workplace	33.5

Where High-Earning Americans Spend Their Time

Top 25% of earners (>= $20.92/hour)

Location	% of Time
Car/Truck/Motorcycle (Driver)	7.8
Other Place	3.8
Someone Else's Home	3.5
Restaurant or Bar	1.9
Outdoors Away from Home	1.5
Other Store/Mall	1.4
Car/Truck/Motorcycle (Passenger)	1.0
Grocery Store	0.7

```
# Question 4: What are the most common activities reported by Ameri

# activities fall under categories like "Relaxing" or "Arts and ent

q4_table <- act_df |>
  filter(activity_location %in% c("Bus", "Subway/Train", "Boat/Ferr
  filter(!str_starts(level1_code, "18")) |>
  left_join(activities_df |> select(code_level3, task_level3),
            by = c("level3_code" = "code_level3")) |>
  filter(!is.na(task_level3)) |>
  filter(!str_detect(tolower(task_level3),
                     "boating|fishing|missing|insufficient|n\\.e\\.
  group_by(task_level3) |>
```

```
  summarise(n_reports = n(), .groups = "drop") |>
  arrange(desc(n_reports)) |>
  head(10)

q4_table |>
  gt() |>
  cols_label(task_level3 = "Activity", n_reports = "Times Reported"
  tab_header(title = "Most Common Activities on Public Transit",
             subtitle = "What Americans do while on a bus, train, s
```

Most Common Activities on Public Transit

What Americans do while on a bus, train, subway, boat, or ferry

Activity	Times Reported
Work, main job	117
Eating and drinking	70
Socializing and communicating with others	40
Household and personal organization and planning	26
Other Arts and entertainment	12
Relaxing, thinking	12
Shopping, except groceries, food, and gas	12
Work, other job(s)	11

Most Common Activities on Public Transit

What Americans do while on a bus, train, subway, boat, or ferry

Activity	Times Reported
Reading for personal interest	10
Television and movies (not religious)	10

```
# Question 5: How much time per day do Americans spend on your four

# Identify codes for four chosen activities
fav_activities <- activities_df |>
  filter(
    str_detect(tolower(task_level2), "socializ") |
    str_detect(tolower(task_level2), "participating in sports|exerc
    str_detect(tolower(task_level2), "television") |
    str_detect(tolower(task_level2), "eating and drinking")
  ) |>
  select(code_level3, task_level2) |>
  # Simplify labels
  mutate(activity = case_when(
    str_detect(tolower(task_level2), "socializ")   ~ "Socializing",
    str_detect(tolower(task_level2), "sports|exercise") ~ "Sports/E
    str_detect(tolower(task_level2), "television") ~ "Watching TV",
    str_detect(tolower(task_level2), "eating")     ~ "Eating & Drin
  ))

q5_table <- act_df |>
  inner_join(fav_activities |> select(code_level3, activity),
             by = c("level3_code" = "code_level3")) |>
  group_by(participant_id, activity) |>
  summarise(total_min = sum(time_spent_min), .groups = "drop") |>
```

```
  right_join(resp_full |> select(participant_id, survey_weight, age
             by = "participant_id") |>
  filter(!is.na(activity), !is.na(age)) |>
  mutate(
    total_min = replace_na(total_min, 0),
    age_group = cut(age, breaks = c(14, 25, 35, 45, 55, 65, 85),
                    labels = c("15-25", "26-35", "36-45", "46-55",
  ) |>
  group_by(age_group, activity) |>
  summarise(
    avg_hours = round(weighted.mean(total_min, survey_weight, na.rm
    .groups = "drop"
  ) |>
  pivot_wider(names_from = activity, values_from = avg_hours)

q5_table |>
  gt() |>
  cols_label(age_group = "Age Group") |>
  tab_spanner(columns = -age_group, label = "Avg. Daily Hours") |>
  tab_header(title = "Time Spent on Favorite Activities by Age Grou
```

Time Spent on Favorite Activities by Age Group

	Avg. Daily Hours		
Age Group	Eating & Drinking	Socializing	Sports/Exercise
15-25	1.15	2.08	2.22
26-35	1.24	2.00	1.74
36-45	1.22	1.82	1.65

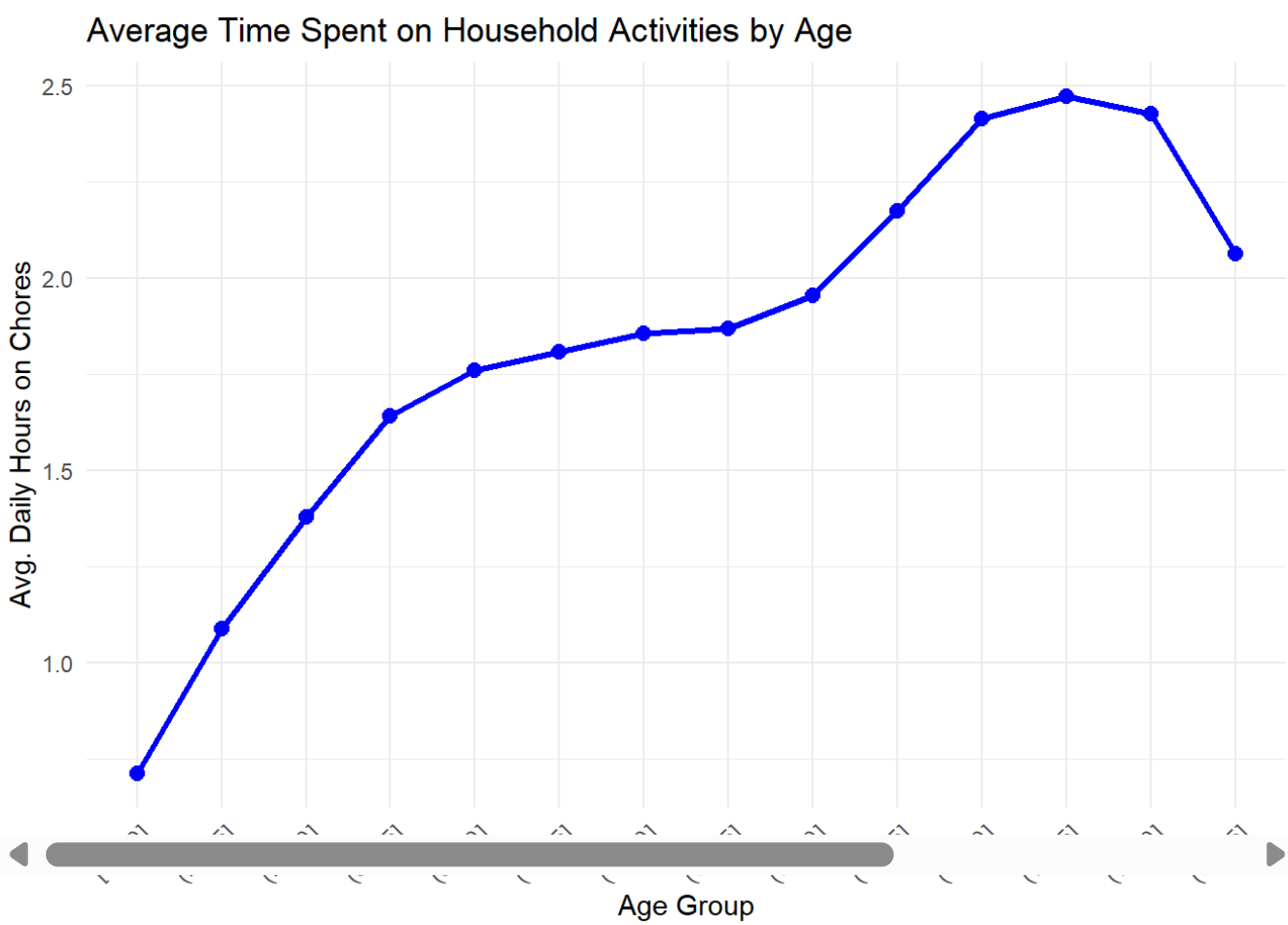
Time Spent on Favorite Activities by Age Group

	Avg. Daily Hours		
Age Group	Eating & Drinking	Socializing	Sports/Exercise
46-55	1.24	1.84	1.56
56-65	1.32	1.90	1.58
66-85	1.44	1.96	1.47

Task 6: Exploratory Questions - Graph Answers

```
# Question 1: How does the amount of time spent on household activi

# Household activities = level 1 code "020000" (Household Activitie
household_codes <- activities_df |>
  filter(code_level1 == "020000") |>
  pull(code_level3)

q6_q1 <- act_df |>
  filter(level3_code %in% household_codes) |>
  group_by(participant_id) |>
  summarise(hh_min = sum(time_spent_min)) |>
  right_join(resp_full |> select(participant_id, survey_weight, age
             by = "participant_id") |>
  mutate(hh_min = replace_na(hh_min, 0)) |>
  filter(!is.na(age)) |>
  mutate(age_group = cut(age,
                         breaks = seq(15, 85, by = 5),
```

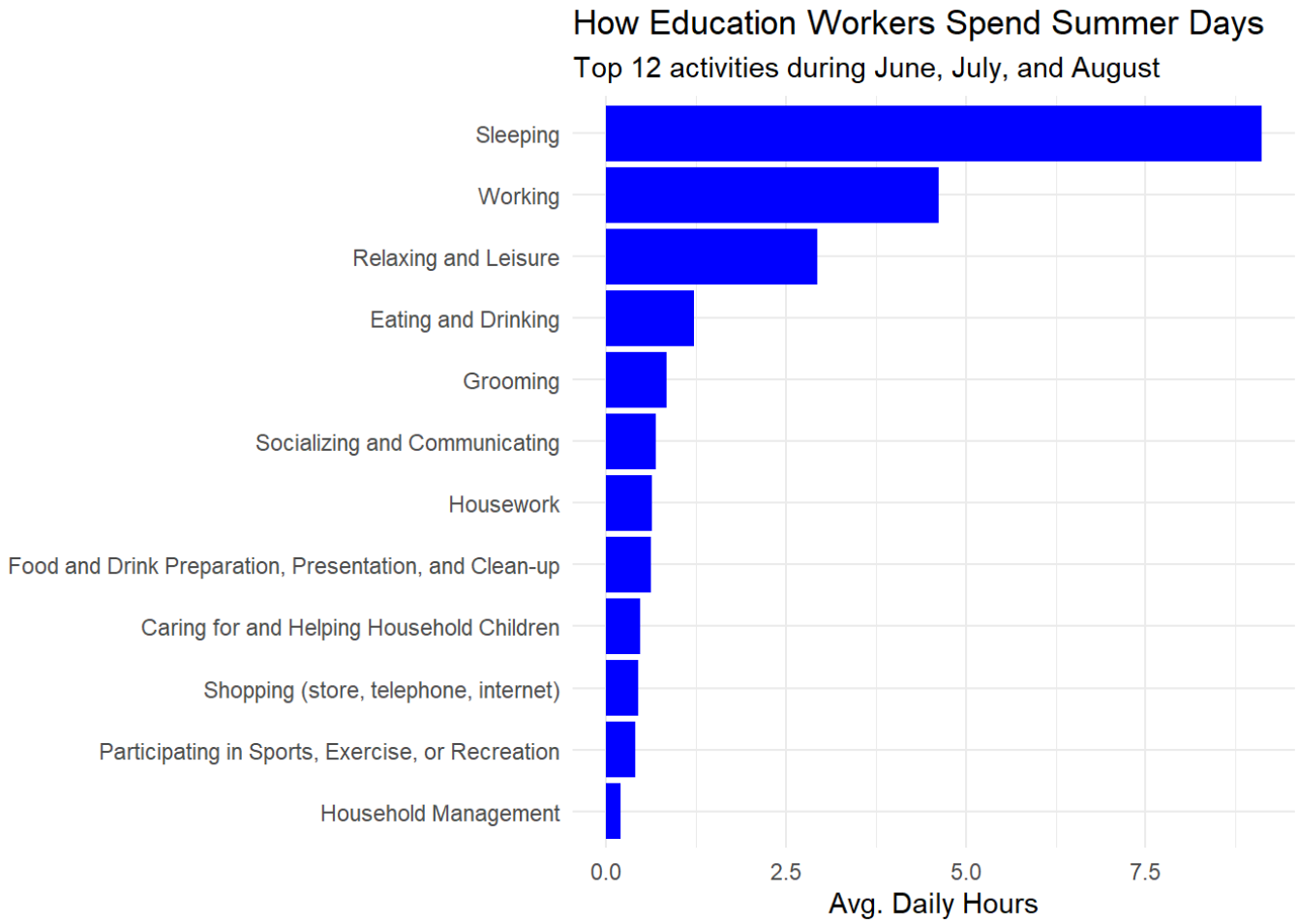
```
                          include.lowest = TRUE)) |>
  filter(!is.na(age_group)) |>
  group_by(age_group) |>
  summarise(avg_hours = weighted.mean(hh_min, survey_weight, na.rm =
            .groups = "drop")

ggplot(q6_q1, aes(x = age_group, y = avg_hours, group = 1)) +
  geom_line(linewidth = 1.2, color = "blue") +
  geom_point(size = 2.5, color = "blue") +
  theme_minimal() +
  labs(title = "Average Time Spent on Household Activities by Age",
       x = "Age Group",
       y = "Avg. Daily Hours on Chores") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



1. Time spent on household activities rises steadily with age for the most part, peaking around the 65–75 age range when respondents are typically retired and have more time to themselves. After that peak, it starts to dip likely because they are getting too old to do the activities. Younger Americans in their teens and twenties spend relatively little time on household tasks, about an hour or less per day. Retirees may spend well over two hours on activities like cooking, cleaning, and home maintenance.

```
# Question 2: What are the most popular activities for Americans em

edu_summer_ids <- resp_full |>
  filter(industry_code >= 7860 & industry_code <= 8192,
         survey_month %in% c(6, 7, 8)) |>
  select(participant_id, survey_weight)

q6_q2 <- act_df |>
  inner_join(edu_summer_ids, by = "participant_id") |>
  filter(!str_starts(level1_code, "18")) |>  # exclude travel
  left_join(activities_df |> select(code_level3, task_level2),
            by = c("level3_code" = "code_level3")) |>
  filter(!is.na(task_level2)) |>
  group_by(task_level2) |>
  summarise(
    total_weighted_min = sum(time_spent_min * survey_weight),
    .groups = "drop"
  ) |>
  mutate(avg_hours = total_weighted_min / sum(total_weighted_min) *
  arrange(desc(avg_hours)) |>
  head(12)

ggplot(q6_q2, aes(x = reorder(task_level2, avg_hours), y = avg_hour
  geom_col(fill = "blue") +
  coord_flip() +
  theme_minimal() +
```

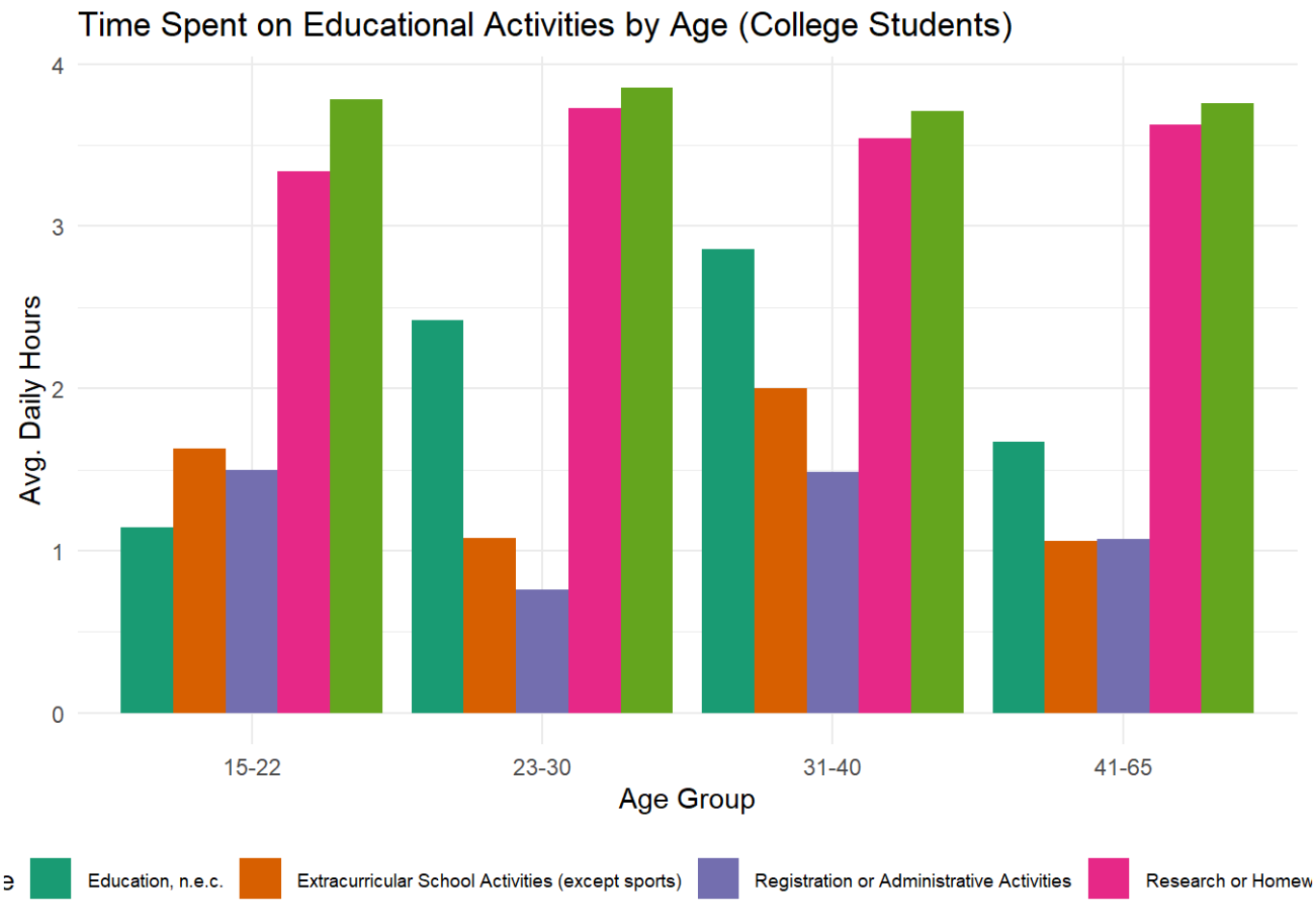
```
  labs(title = "How Education Workers Spend Summer Days",
       subtitle = "Top 12 activities during June, July, and August",
       x = NULL,
       y = "Avg. Daily Hours")
```



2. During the summer months, education workers spend most of their time sleeping, working (possibly preparation for the next school year, or working on a side project), and relaxing. They also spend plenty of time going out to eat, hanging out with their friends, and taking care of the house.This makes sense because they have the whole day off and they're able to do more activities because they aren't teaching over the summers.

```
# Question 3:  Among respondents who report being enrolled in a high
```

```
# Education activities = level 1 code "060000"
edu_codes <- activities_df |>
  filter(code_level1 == "060000") |>
  pull(code_level3)

# Get only respondents enrolled in college
college_ids <- resp_full |>
  filter(college_enrolled == "Yes") |>
  select(participant_id, survey_weight, age)

q6_q3 <- act_df |>
  filter(level3_code %in% edu_codes) |>
  left_join(activities_df |> select(code_level3, task_level2),
            by = c("level3_code" = "code_level3")) |>
  group_by(participant_id, task_level2) |>
  summarise(edu_min = sum(time_spent_min), .groups = "drop") |>
  right_join(college_ids, by = "participant_id") |>
  filter(!is.na(task_level2)) |>
  mutate(edu_min = replace_na(edu_min, 0)) |>
  mutate(age_group = cut(age,
                         breaks = c(14, 22, 30, 40, 65, 85),
                         labels = c("15-22", "23-30", "31-40", "41-
  filter(!is.na(age_group)) |>
  group_by(age_group, task_level2) |>
  summarise(avg_hours = weighted.mean(edu_min, survey_weight, na.rm
            .groups = "drop")

ggplot(q6_q3, aes(x = age_group, y = avg_hours, fill = task_level2)
  geom_col(position = "dodge") +
  theme_minimal() +
  labs(title = "Time Spent on Educational Activities by Age (Colleg
       x = "Age Group",
       y = "Avg. Daily Hours") +
  scale_fill_brewer(type = "qual", palette = 2, name = "Activity Ty
  theme(legend.position = "bottom",
        legend.text = element_text(size = 7))
```

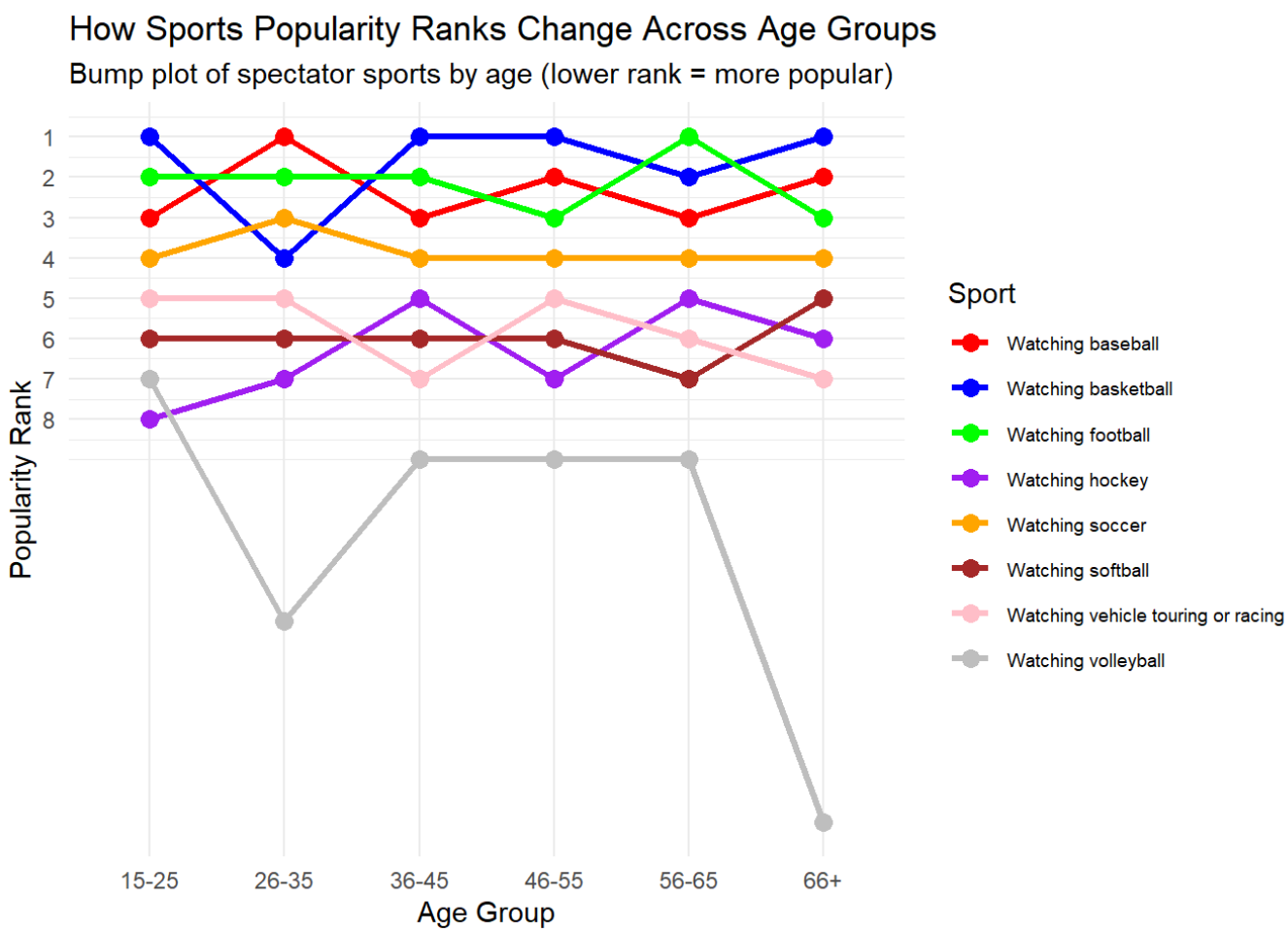


3. The data provides some support for the idea that older students take their education more seriously. While traditional-age students (15–22) spend the most of their time in classes which makes sense since many attend full time. Older students (30+) tend to allocate proportionally more of their education time to homework and research activities. This can be because a lot of students who are 30+ can be going for a phd, an mba, or maybe they're in medical school. This suggests that while younger students may be physically present in class more often, non-traditional students may engage more deeply with independent study, giving some credibility to the "wasted on the young" remark.

```
# Question 4: Use a bump plot to show how the relative interest in

# Get all "Attending Sports or Recreational Events" level 3 codes
```

```
sports_codes <- activities_df |>
  filter(task_level2 == "Attending Sports or Recreational Events")
  select(code_level3, sport = task_level3) |>
  filter(!is.na(sport)) |>
  # Remove generic "Other" category
  filter(!str_detect(tolower(sport), "other|n\\.e\\.c"))

q6_q4 <- act_df |>
  inner_join(sports_codes, by = c("level3_code" = "code_level3")) |
  left_join(resp_full |> select(participant_id, survey_weight, age)
            by = "participant_id") |>
  filter(!is.na(age)) |>
  mutate(age_group = cut(age,
                         breaks = c(14, 25, 35, 45, 55, 65, 85),
                         labels = c("15-25", "26-35", "36-45", "46-
  filter(!is.na(age_group)) |>
  group_by(age_group, sport) |>
  summarise(n = n(), .groups = "drop") |>
  group_by(age_group) |>
  mutate(rank = rank(-n, ties.method = "first")) |>
  ungroup()

# Keep only sports that rank top 7 in at least one age group
top_sports <- q6_q4 |> filter(rank <= 7) |> pull(sport) |> unique()
q6_q4 <- q6_q4 |> filter(sport %in% top_sports)

ggplot(q6_q4, aes(x = age_group, y = rank, color = sport, group = s
  geom_line(linewidth = 1.2) +
  geom_point(size = 3) +
  scale_y_reverse(breaks = 1:8) +
  theme_minimal() +
  labs(title = "How Sports Popularity Ranks Change Across Age Group
       subtitle = "Bump plot of spectator sports by age (lower rank
       x = "Age Group",
       y = "Popularity Rank") +
  theme(legend.position = "right",
```

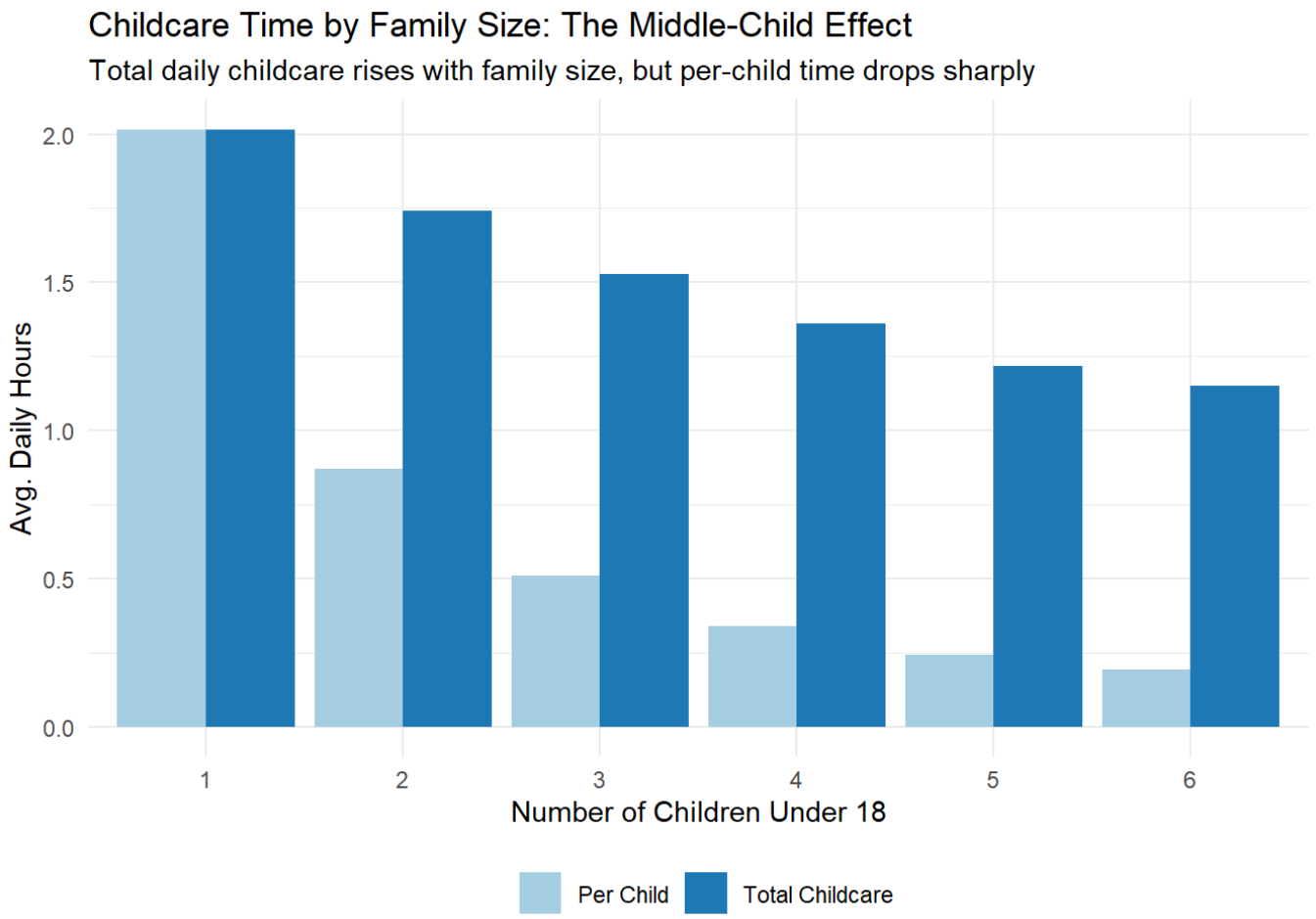
```
        legend.text = element_text(size = 7)) +
  scale_color_manual(values = c("red", "blue", "green", "purple",
                                "orange", "brown", "pink", "grey"),
                     name = "Sport")
```



4. The bump plot shows clear generational differences in spectator sports preferences. Football and basketball are more popular among younger age groups, while golf and horse racing gain relative popularity among older Americans. Baseball remains fairly stable across all ages. These shifts likely reflect both cultural trends and the accessibility of different sports for older viewers

```
# Question 5: How does the amount of time parents spend per child cl

# Childcare codes: level2 030100, 030200, 030300
```

```
childcare_codes <- c("030100", "030200", "030300")

parents <- resp_full |>
  filter(n_children_under18 >= 1) |>
  select(participant_id, survey_weight, n_children_under18)

q6_q5 <- act_df |>
  filter(level2_code %in% childcare_codes) |>
  group_by(participant_id) |>
  summarise(care_min = sum(time_spent_min)) |>
  right_join(parents, by = "participant_id") |>
  mutate(care_min = replace_na(care_min, 0),
         # Per-child time
         care_per_child_min = care_min / n_children_under18) |>
  group_by(n_children_under18) |>
  summarise(
    avg_total_hours = weighted.mean(care_min, survey_weight, na.rm =
    avg_per_child_hours = weighted.mean(care_per_child_min, survey_w
    .groups = "drop"
  ) |>
  filter(n_children_under18 <= 6) |>  # drop very small sample size
  pivot_longer(cols = c(avg_total_hours, avg_per_child_hours),
               names_to = "measure",
               values_to = "hours") |>
  mutate(measure = if_else(measure == "avg_total_hours",
                           "Total Childcare", "Per Child"))

ggplot(q6_q5, aes(x = factor(n_children_under18), y = hours, fill =
  geom_col(position = "dodge") +
  theme_minimal() +
  labs(title = "Childcare Time by Family Size: The Middle-Child Effe
       subtitle = "Total daily childcare rises with family size, but
       x = "Number of Children Under 18",
       y = "Avg. Daily Hours") +
  scale_fill_brewer(type = "qual", palette = 3, name = "") +
  theme(legend.position = "bottom")
```



The data clearly supports the “middle-child syndrome” hypothesis. Besides for spending less time with each kid as you have more and more kids, parents also spend less time total with each of their kids. However I think that can be explained that as you have more kids, more time has passed which means some kids are now in school. Also now more money may be saved and the parents may get a nanny for the babies. A single child household may see two hours of dedicated parental attention per day, while each child in a four-child household receives closer to 20 minutes. Parents do scale up their effort, but not nearly enough to maintain the same per-child investment, meaning that later children inevitably receive less individual attention.

Task 7: Time Use Categories for Target Demographics

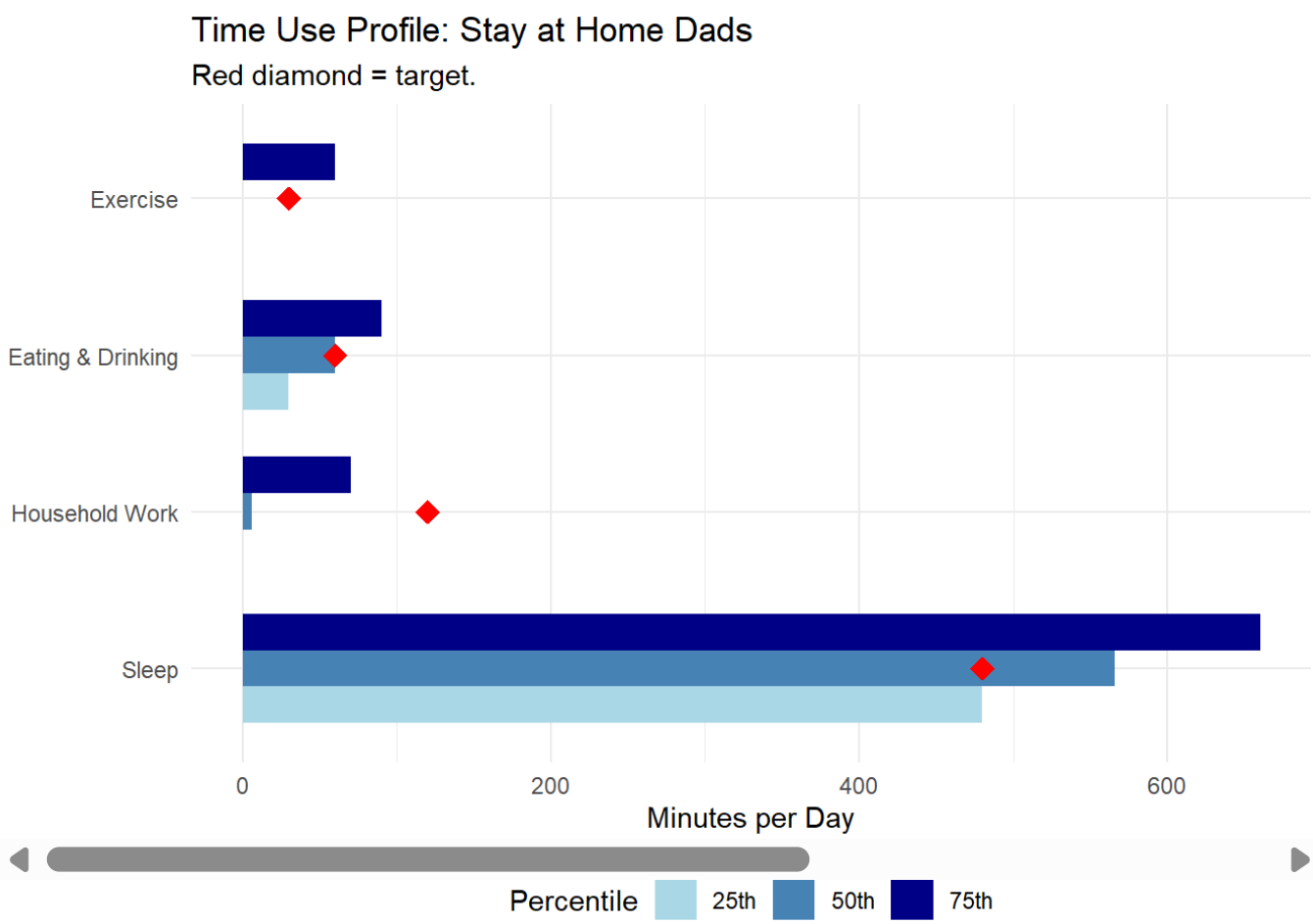
For our time-tracking app, we define three target user personas and analyze their typical time-use patterns using ATUS data. For each, we identify key activities, compute “low” (25th percentile), “medium” (50th), and “high” (75th) usage levels, set a target, and determine what fraction of the demographic hits that target.

```
# Helper plot

# Activity code sets used across all three demos
household_codes <- activities_df |> filter(code_level1 == "020000")
exercise_codes  <- activities_df |> filter(str_detect(tolower(task_
eating_codes    <- activities_df |> filter(str_detect(tolower(task_
sleep_codes     <- c("010101")
social_codes    <- activities_df |> filter(str_detect(tolower(task_
work_codes      <- activities_df |> filter(task_level2 == "Working"
class_codes     <- activities_df |> filter(str_detect(tolower(task_
homework_codes  <- activities_df |> filter(str_detect(tolower(task_
childcare_codes <- activities_df |> filter(code_level2 %in% c("0301
travel_codes    <- activities_df |> filter(code_level1 == "180000")
leisure_codes   <- activities_df |> filter(code_level1 == "120000")

# Helper: filter act_df for specific codes within a demographic
get_acts <- function(codes, demo) {
  act_df |> filter(level3_code %in% codes) |> semi_join(demo, by =
}

# Helper: compute weighted percentiles and return plottable data fr
compute_time_plot <- function(demo_acts, demo_resp, label, target_m
  pp <- demo_acts |>
    group_by(participant_id) |>
    summarise(total_min = sum(time_spent_min), .groups = "drop") |>
```

```
    right_join(demo_resp |> select(participant_id, survey_weight), 
    mutate(total_min = replace_na(total_min, 0))

  ord <- pp |> arrange(total_min)
  cw <- cumsum(ord$survey_weight) / sum(ord$survey_weight)

  tibble(
    activity = label,
    percentile = c("25th", "50th", "75th"),
    minutes = c(
      ord$total_min[which(cw >= 0.25)[1]],
      ord$total_min[which(cw >= 0.50)[1]],
      ord$total_min[which(cw >= 0.75)[1]]
    ),
    target = target_min
  )
}
```

```
# Stay at home dads

sahd <- resp_full |> filter(sex == "M", n_children_under18 >= 1, em

sahd_plot <- bind_rows(
  compute_time_plot(get_acts(household_codes, sahd), sahd, "Househo
  compute_time_plot(get_acts(exercise_codes, sahd),  sahd, "Exercis
  compute_time_plot(get_acts(eating_codes, sahd),    sahd, "Eating 
  compute_time_plot(get_acts(sleep_codes, sahd),     sahd, "Sleep",
) |>
  mutate(percentile = factor(percentile, levels = c("25th", "50th",
         activity = factor(activity, levels = c("Sleep", "Household

ggplot(sahd_plot, aes(x = activity, y = minutes, fill = percentile)
  geom_col(position = "dodge", width = 0.7) +
  geom_point(aes(x = activity, y = target),
             inherit.aes = FALSE,
```

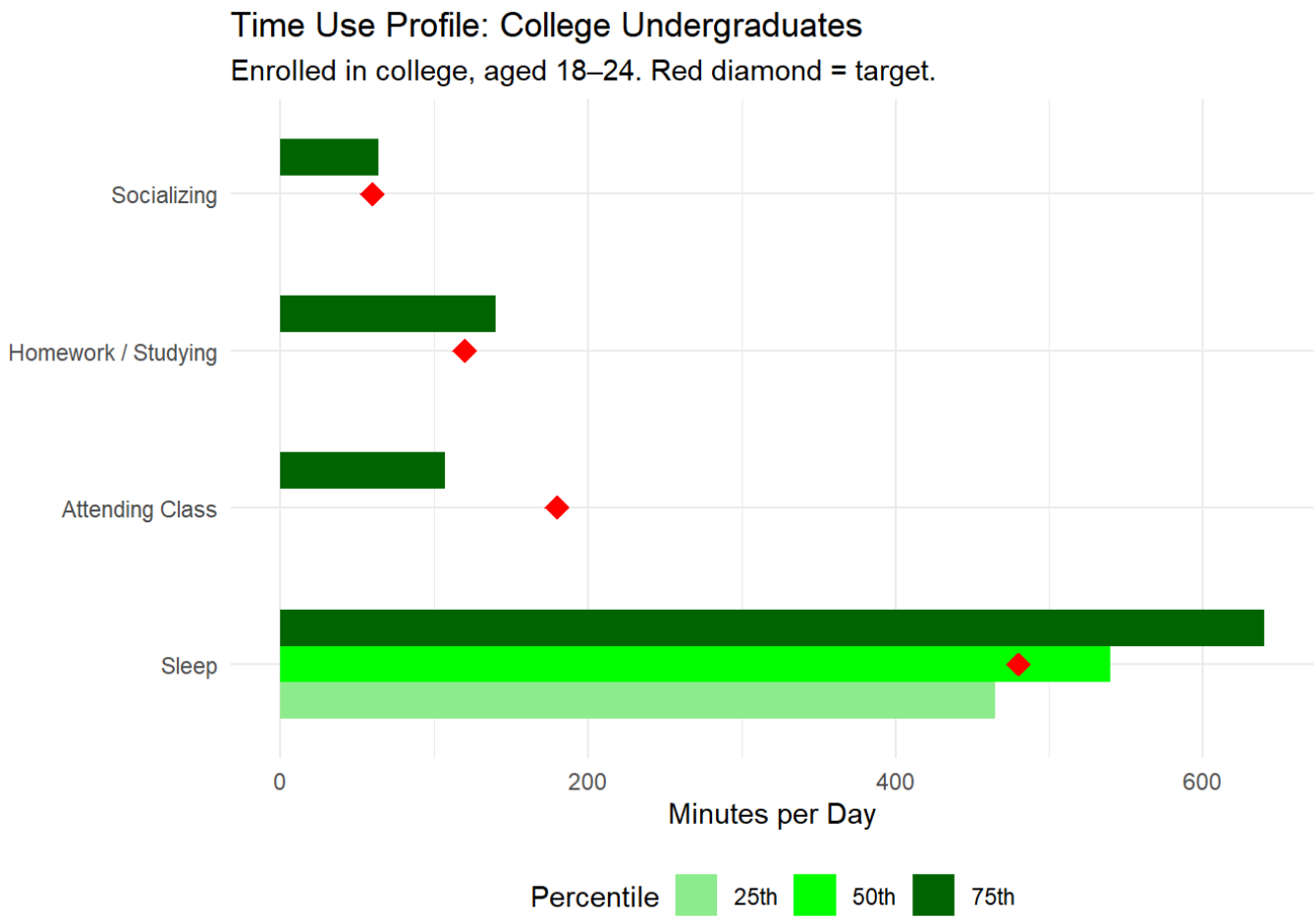
```
                shape = 18, size = 4, color = "red") +
  coord_flip() +
  scale_fill_manual(values = c("25th" = "lightblue", "50th" = "stee
                    name = "Percentile") +
  theme_minimal() +
  labs(title = "Time Use Profile: Stay at Home Dads",
       subtitle = "Red diamond = target.",
       x = NULL,
       y = "Minutes per Day") +
  theme(legend.position = "bottom")
```



Stay at home dads show a clear split between high-variation and low-variation activities. Sleep dominates the day at a median of nearly 9.5 hours, well above the 8-hour target. Household work and eating show moderate engagement, with the 75th percentile dads putting in over an

hour on chores. Exercise is the most skewed: the majority of SAHDs get none on a given day, but the top quartile manages a full hour — suggesting the app could use peer comparison to motivate the less active dads.

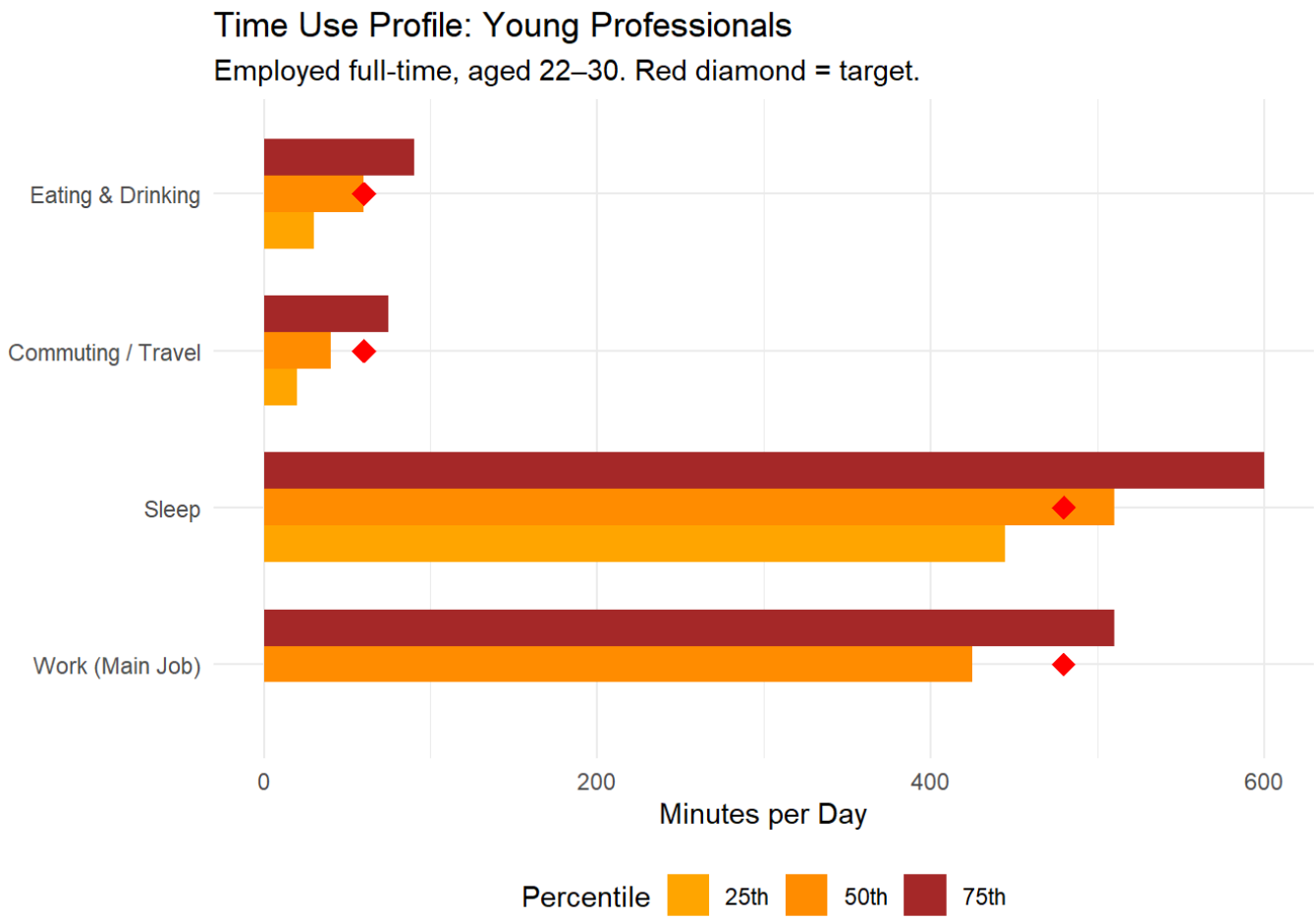
```
# Undergrads

undergrads <- resp_full |> filter(college_enrolled == "Yes", age >=

ug_plot <- bind_rows(
  compute_time_plot(get_acts(class_codes, undergrads),    undergrad
  compute_time_plot(get_acts(homework_codes, undergrads), undergrad
  compute_time_plot(get_acts(social_codes, undergrads),   undergrad
  compute_time_plot(get_acts(sleep_codes, undergrads),    undergrad
) |>
  mutate(percentile = factor(percentile, levels = c("25th", "50th",
         activity = factor(activity, levels = c("Sleep", "Attending

ggplot(ug_plot, aes(x = activity, y = minutes, fill = percentile)) +
  geom_col(position = "dodge", width = 0.7) +
  geom_point(aes(x = activity, y = target),
             inherit.aes = FALSE,
             shape = 18, size = 4, color = "red") +
  coord_flip() +
  scale_fill_manual(values = c("25th" = "lightgreen", "50th" = "gre
                    name = "Percentile") +
  theme_minimal() +
  labs(title = "Time Use Profile: College Undergraduates",
       subtitle = "Enrolled in college, aged 18–24. Red diamond = t
       x = NULL,
       y = "Minutes per Day") +
  theme(legend.position = "bottom")
```



The undergraduate chart shows a big contrast between sleep and academic activities. While students comfortably exceed the 8 hour sleep target at the median, class attendance and homework are very right skewed: most students log zero hours of studying, but those who do study put in over two hours. Socializing follows a similar pattern. For the app, this means the biggest wins come from encouraging consistent daily study habits - even 30 minutes of homework on an off day, or a 30 minute walk with a friend would put a student above the median.

```
# Young professionals

young_pros <- resp_full |> filter(age >= 22, age <= 30, employment_

travel_codes <- activities_df |> filter(code_level1 == "180000") |>
```

```
yp_plot <- bind_rows(
  compute_time_plot(get_acts(work_codes, young_pros),   young_pros,
  compute_time_plot(get_acts(sleep_codes, young_pros),  young_pros,
  compute_time_plot(get_acts(eating_codes, young_pros), young_pros,
  compute_time_plot(get_acts(travel_codes, young_pros), young_pros,
) |>
  mutate(percentile = factor(percentile, levels = c("25th", "50th",
         activity = factor(activity, levels = c("Work (Main Job)",

ggplot(yp_plot, aes(x = activity, y = minutes, fill = percentile)) +
  geom_col(position = "dodge", width = 0.7) +
  geom_point(aes(x = activity, y = target),
             inherit.aes = FALSE,
             shape = 18, size = 4, color = "red") +
  coord_flip() +
  scale_fill_manual(values = c("25th" = "orange", "50th" = "darkorai
                    name = "Percentile") +
  theme_minimal() +
  labs(title = "Time Use Profile: Young Professionals",
       subtitle = "Employed full-time, aged 22–30. Red diamond = tai
       x = NULL,
       y = "Minutes per Day") +
  theme(legend.position = "bottom")
```



Young professionals split their day mostly between work and sleep, with the median getting about 7 hours at work and 8–9 hours sleeping. Commuting and travel takes up a notable part of the day, with the top quartile spending over an hour commuting. Eating and drinking is consistent but modest. The target diamonds reveal that most young professionals fall just short of a full 8-hour workday and a full hour of eating. The app could frame this as “You took a real lunch break today — only 40% of professionals your age do that!”

AI Usage Statement